

AI-Powered Fake News Detection Using Text Classification

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The Threat of Fake News

Unprecedented Misinformation

The rapid growth of digital media platforms has significantly accelerated the speed of information dissemination, allowing fabricated content to manipulate public opinion globally.

Fake news severely impacts political decisions, financial markets, and public health. With millions of articles generated daily, traditional manual verification is entirely unscalable.

Artificial Intelligence and NLP offer powerful, automated solutions to identify discriminative textual patterns associated with misleading content.



Project Objectives



NLP Preprocessing

Develop a robust text cleaning pipeline to standardize data, remove noise, and eliminate URLs, tags, and stopwords to significantly improve dataset quality.



Feature Engineering

Transform raw text into mathematical vectors using Bag-of-Words (BoW), Term Frequency-Inverse Document Frequency (TF-IDF), and semantic Word2Vec models.



Model Evaluation

Train and rigorously compare multiple machine learning algorithms (LR, KNN, Random Forest, MLP) to identify the highest performing classifier.

Dataset Overview

44,898

Total News Articles

Kaggle Benchmark Dataset

The models were trained on a highly balanced, publicly available dataset comprising authentic news from verified sources and fabricated content from unreliable domains.

The dataset was meticulously merged, labeled (0 for Fake, 1 for Real), randomly shuffled, and subjected to a strict 80:20 split, yielding **35,918** training samples and **8,980** testing samples.

Detection Methodology

2. Tokenization

Segmenting cleaned sentences into individual word tokens and filtering out universally common stopwords using NLTK to reduce dimensionality.

4. Evaluation

Training classifiers and evaluating via Confusion Matrices, Accuracy, Precision, Recall, and F1-Scores on the reserved testing set.

1. Data Cleaning

Aggressive lowercase conversion, removal of hyperlinks, HTML tags, punctuation, and noisy characters to standardize raw textual input.

3. Vectorization

Mapping textual tokens into dense or sparse continuous feature vectors via TF-IDF and continuous semantic neural embeddings (Word2Vec).

Feature Engineering

TF-IDF Representation

Extends simple frequency counts by penalizing universally common terms. It assigns larger weights to highly discriminative words unique to specific documents.

$$\text{IDF}(t) = \log \frac{N}{\text{DF}(t)}$$

Word2Vec Embeddings

A sophisticated neural embedding technique that converts words into dense, continuous vectors. It effectively captures deep semantic similarity between words based on contextual usage patterns.

Machine Learning Algorithms

Logistic Regression

A highly efficient, interpretable parametric model. It leverages the sigmoid activation function to compute class probabilities, performing exceptionally well on sparse, high-dimensional TF-IDF vectors.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Random Forest

An advanced ensemble technique combining multiple decision trees. Through Bootstrap Aggregation (Bagging), it dramatically reduces prediction variance and effectively handles noisy datasets.

K-Nearest Neighbors

A non-parametric model that classifies articles based on their closest neighbors in the semantic feature space, generally leveraging Euclidean distance across embeddings.

$$d = \sqrt{\sum_{k=1}^n x_i - x_j^2}$$

Neural Network (MLP)

A Multi-Layer Perceptron utilizing interconnected layers and backpropagation to model highly complex, non-linear textual relationships.

Experimental Results



Observation: All models achieved accuracy exceeding 97%. Random Forest delivered the highest overall performance across all evaluation metrics, demonstrating near-perfect classification capabilities.

Discussion & Conclusion

The Power of Classical ML

The experimental outcomes unequivocally demonstrate that properly engineered traditional machine learning pipelines remain highly competitive for text classification tasks.

The combination of rigorous preprocessing and discriminative feature extraction (such as TF-IDF) allowed even simplistic models like Logistic Regression to achieve near-optimal results.

These classical methods offer a massive advantage: they achieve high accuracy while requiring **significantly lower computational resources** compared to intensive deep learning transformer architectures.

